

Perfectly Still: Stability and Policy Response Capacity

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Abstract

Policy responses depend on the units, histories, contracts, and configurations through which an intervention changes behavior. I define policy response capacity as a finite-horizon policy-to-target derivative for continuous instruments and as a finite contrast for discrete interventions. Systems sharing a stable baseline law can have response ranks from zero to full row rank, and attraction to a zero-response state supplies a time bound for entering any stated capacity region. In 186,893 active-phase decision points from the myBPmyLife micro-randomized trial, moving assigned notification history from its 10th to its 90th percentile over 14 days reduces the current diet-notification response by 1.75 percentage points. Across four exploratory engagement targets and two notification types, the standardized response-change operators at seven and 14 days have leading singular values 0.142 and 0.136; simultaneous five-window lower endpoints are 0.023 and 0.017. The 14-day lower-rank-uniform tests reject rank zero and retain rank at most one. A separate 1,933-person factorial experiment shows that randomized app limits change the use response to a temporary bonus during its payment window. These history and configuration effects enter response transitions and a planner's finite assigned actions.

JEL codes: C23, C61, D12, I15, I18. *Keywords:* policy response capacity, randomized policy history, response matrix, finite policy actions, factorial experiment.

1. Introduction

A policy moves an economy through households, firms, contracts, and accounts that receive an intervention and change their behavior. Filing, enrollment, purchase, exhaustion, refinancing, and exit determine which units respond today. Current intervention changes those states before the next policy arrives. The multiplier faced by the next decision maker therefore evolves with past policy.

The paper follows this evolution through an endpoint map from a policy sequence to a vector of targets. Its derivative with respect to continuous instruments is a response matrix, and its column space contains the target directions generated locally by those instruments. A discrete policy supplies a finite contrast. I call these objects policy response capacity and carry them as part of the economic state. Public-policy research uses policy capacity for the analytical, operational, and political competences held by policy actors (Wu et al. 2015). Policy response capacity here refers to target changes induced by a stated intervention over a stated horizon.

The unit representation separates attachment to a policy system from response conditional on attachment. For observed binary attachment, the change in the response matrix decomposes into three terms: units entering between dates, units exiting between dates, and response changes among units present at both dates. The decomposition uses joint transitions. Marginal participation rates identify net changes, while joint histories identify the gross-flow terms. Differentiating the response matrix with respect to an earlier intervention produces a response-of-response estimand.

Policy design depends on the response additions available to the decision maker. Let V be the column space of the current matrix and let W collect target directions required by an objective. With unrestricted additions, the number of missing quotient directions is $\dim W - \dim(W \cap V)$. Administrative and technological restrictions supply a dictionary of feasible response additions. The minimum number and cost of additions depend on which projected dictionary vectors span the required directions. A finite assignment design supplies an action set $\mathcal{K}$. The planner evaluates every point in $\mathcal{K}$; population mixtures generate $\text{conv}(\mathcal{K})$; minimization over $\text{span}(\mathcal{K})$ gives a relaxation whose value is attained by the experiment exactly when a closest point belongs to the attainable set.

Two results connect response capacity to baseline dynamics. Holding the transition matrix and target map fixed, systems with the same stable baseline law admit response ranks from zero to full row rank. Along an exponentially attracting path, Lipschitz continuity of response strength transfers the convergence rate into a bound on the time required to enter any stated low-capacity region. Stability describes the baseline return path, while the response matrix describes target movements generated by the intervention set.

A contemporaneous model derives the interaction between an incentive and a configuration. Desired use falls under a use-reduction bonus, and an effective cap truncates

realized use. The change in bonus response equals the length of the cap change that crosses the interval between incentivized and unincentivized use; averaging gives a population overlap formula for soft caps and heterogeneous use. A two-date adoption model connects the response-of-response derivative to household decisions. A subsidy in the first date lowers the adoption threshold. Marginal early adopters enter the second date with product experience and one unit already owned. Experience raises valuation of a later unit, while ownership lowers its marginal value. The change in later response capacity equals the density of marginal early adopters multiplied by the change in their response to the later offer.

The first application uses the public myBPmyLife micro-randomized trial (Golbus et al. 2026). The 287-person core cohort received 187,517 randomization points over six months. Four daily draws assigned a diet notification with probability 0.125, an activity notification with probability 0.125, and the control condition with probability 0.75. The analysis uses 186,893 active-phase points and asks how assigned notification history changes the response to a current diet notification. The source article labels 60-minute app engagement an exploratory post hoc endpoint; the outcome retains that status here.

The same decision panel supplies a passive state transition and a randomized response-history contrast. Under the control draw, app use during the next 60 minutes equals 4.22 percent after a zero 30-minute pre-state and 19.76 percent after a one pre-state. Their difference is 0.155, and the corresponding two-state iteration has a fixed point of 5.00 percent. At mean randomized history following a zero pre-state, a current diet notification raises 60-minute app use by about 5 percentage points. Moving from the 10th to the 90th percentile of assigned notification history over the preceding 14 days reduces that response by 1.75 percentage points with a participant-clustered standard error of 0.60. Gaussian cluster-multiplier adjustment gives $p = 0.0136$ across five displayed history windows and $p = 0.0766$ across the 40 coefficients screened over five windows, four outcomes, and two notification types. A future-history interaction gives $p = 0.206$.

The 40 coefficients also form five response-change operators with four engagement targets and two current notification types. At seven and 14 days, their leading standardized singular values are 0.142 and 0.136. A five-window simultaneous operator-norm band gives positive lower endpoints of 0.023 and 0.017, and the adjusted rank-zero values are 0.012 and 0.017. At 14 days, a lower-rank-uniform test retains rank at most one with $p = 0.577$. This retrospective operator diagnostic supports at least one history-sensitive response direction and leaves a second direction unresolved.

The response equation includes app use measured before current assignment and interactions between that pre-state and both current notification assignments. The history coefficients therefore compare current responses at the same recorded state. Prior assignment changes a current policy multiplier remaining after current-state response

heterogeneity enters the equation. The analysis record preserves the candidate screen, both simultaneous families, the future-history audit, the pinned source commit, and the source endpoint classification.

The second application uses the 1,933-person factorial experiment in Allcott et al. (2022). Participants received independent assignments to a temporary bonus for lower digital use and to an app-limit configuration. During the payment period, the bonus reduces use by 62.9 minutes per day in the limit-control group and by 51.5 minutes under a limit assignment. The 11.4-minute interaction has standard error 6.2. An exploratory payment-window contrast equals 14.1 minutes; Gaussian and conditional re-randomization family adjustments give 0.020 and 0.0216. A four-action planner maps the assigned combinations into daily reduction targets and implementation-cost boundaries.

Registered experiments on distracted driving and mobile-phone goals carry configuration evidence to separate populations (Delgado et al. 2024; Sachdeva et al. 2026). Two public panels extend the history and reach margins. An early bed-net subsidy raises purchase from a common offer one year later by 7.2 percentage points in Dupas (2014). In the StayWell panel, 92.7 percent of a fixed 177-person cohort supplies a first response within 50 prompts (Arévalo Avalos et al. 2024). Each application retains its assignment design, target, horizon, and support.

The targets-and-instruments program studies mappings from instruments into targets (Tinbergen 1952). Field experiments with several instruments and targets estimate how treatment combinations span policy objectives (Hahn et al. 2016), and policy-interaction research measures complementarity and substitution across instruments (Pless 2026). Micro-randomized trials estimate time-varying proximal effects under sequential assignment (Boruvka et al. 2018). Factorial inference uses the saturated main-and-interaction specification (Muralidharan et al. 2025). Inference after selection among estimated options motivates simultaneous coverage over declared finite families (Andrews et al. 2024). Dynamic controllability extends instrument mappings across dates (Buiter and Gersovitz 1981). State-dependent multiplier research estimates policy responses across slack, credit conditions, balance sheets, and histories (Auerbach and Gorodnichenko 2012; Ramey and Zubairy 2018; Eichenbaum et al. 2022). A transition law for the response matrix links these strands to randomized unit histories and configurations, while the empirical choice set remains within the actions generated by each design.

Research on program take-up estimates how information, application assistance, and access change participation (Bhargava and Manoli 2015; Finkelstein and Notowidigdo 2019; Deshpande and Li 2019). Research on experience goods and temporary subsidies studies learning, price exposure, and later adoption (Bergemann and Välimäki 2006; Dupas 2014). Allcott et al. (2022) estimate habit formation and self-control from randomized bonuses and limits. The framework places these estimates in a transition system where an

intervention changes current outcomes and the response supplied to another instrument.

The framework defines policy response capacity and follows it from baseline dynamics to feasible response additions and finite assigned actions. Instrument overlap and adoption then supply two mechanisms that move capacity. The identification and application sections connect those mechanisms to sequential-history and factorial designs, finite-family and rank inference, assigned-action choice, and the registered benchmarks. Proofs, data construction, and specification details appear in the appendix.

2. Policy Response Capacity

Let the full state be $s_t = (x_t, r_t)$, where x_t contains the conventional economic state and r_t contains the units and response parameters through which policy operates. For an admissible policy sequence $U = (u_0, \dots, u_{H-1}) \in \mathcal{U}^H$, let $\Phi_H(s, U)$ denote the terminal state and let τ select a q -dimensional target. The endpoint map is

$$(1) \quad \Psi_H(s, U) = \tau(\Phi_H(s, U)).$$

For continuous instruments, the response matrix and response space are

$$(2) \quad M_H(s) = D_U \Psi_H(s, 0), \quad \mathcal{V}_H(s) = \text{Im } M_H(s).$$

The matrix has q rows and mH columns when there are m instruments per date. Its rank records the number of locally reached target directions. Under stated target and instrument units, its singular values record response strength along orthogonal directions.

For a discrete policy $b \in \{0, 1\}$, define the response-capacity contrast

$$(3) \quad K_H(s) = \mathbb{E}[\Psi_H(s, b = 1) - \Psi_H(s, b = 0) \mid s].$$

Equation (3) supplies the experimental counterpart of a derivative. Continuous and discrete instruments can be stacked after their units are reported.

For a finite assignment set $\mathcal{A}$, let $k(a) \in \mathbb{R}^q$ be the target change generated by action a relative to a baseline action, and let $c(a)$ be its activation cost. The attainable response set is $\mathcal{K} = \{k(a) : a \in \mathcal{A}\}$. A planner facing target change z chooses

$$(4) \quad L_{\mathcal{A}}(z) = \min_{a \in \mathcal{A}} \left\{ \frac{1}{2} (z - k(a))' Q (z - k(a)) + c(a) \right\}.$$

PROPOSITION 1 (Finite-action policy choice). *Suppose Q is positive definite. Equation (4) compares the target losses of the attainable actions. If the planner can randomize assignment shares and target changes aggregate linearly, the attainable set becomes $\text{conv}(\mathcal{K})$. With zero activation costs, replacing either set by $\text{span}(\mathcal{K})$ gives a weakly lower loss. Equality holds exactly*

when a closest point in the span belongs to the relevant attainable set.

The finite set applies to mutually exclusive experimental arms. Its convex hull applies to population assignment shares. The span applies to continuously scalable instruments whose admissible coefficients include the projection used by the targeting problem.

DEFINITION 1 (Perfect stillness). *A baseline fixed point s^* is perfectly still for a stated target, horizon, and policy class when it is locally attracting under the baseline law and every response derivative and discrete contrast in that class equals zero at s^* .*

2.1. Units and exact transition accounting

Consider N units with weights $\omega_i > 0$. Let $I_{it} \in \{0, 1\}$ indicate attachment to the policy system and let $B_{it} \in \mathbb{R}^{q \times m}$ be unit i 's target response conditional on attachment. The aggregate response matrix is

$$(5) \quad M_t = \sum_{i=1}^N \omega_i I_{it} B_{it}.$$

Attachment can mean enrollment, an active account, an unexhausted claim, a refinancing incentive, or availability to an offer. The conditional matrix can include direct and equilibrium responses under a stated exposure mapping.

THEOREM 1 (Transition accounting). *The change in equation (5) satisfies*

$$(6) \quad M_{t+1} - M_t = E_t - X_t + S_t,$$

$$(7) \quad E_t = \sum_i \omega_i (1 - I_{it}) I_{i,t+1} B_{i,t+1},$$

$$(8) \quad X_t = \sum_i \omega_i I_{it} (1 - I_{i,t+1}) B_{it},$$

$$(9) \quad S_t = \sum_i \omega_i I_{it} I_{i,t+1} (B_{i,t+1} - B_{it}).$$

The three terms collect entrants, exits, and response changes among units attached at both dates.

The theorem uses the joint transition $(I_{it}, I_{i,t+1})$. When attachment is latent and only marginal probabilities $p_{it} = \Pr(I_{it} = 1)$ are available, a gross-flow decomposition additionally requires the transition probabilities $\pi_{it}^{ab} = \Pr(I_{it} = a, I_{i,t+1} = b)$ and response matrices indexed by transition type. The expected terms then replace the indicators in equation (6) with π^{01} , π^{10} , and π^{11} . Marginal probabilities determine $p_{i,t+1} - p_{it} = \pi_{it}^{01} - \pi_{it}^{10}$ and leave the two gross flows separate only after π^{01} or π^{10} is supplied.

Let an earlier policy a_t change next-period attachment and conditional response. For

the probability representation

$$(10) \quad M_{t+1}(a_t) = \sum_i \omega_i p_{i,t+1}(a_t) B_{i,t+1}(a_t),$$

differentiation gives

$$(11) \quad D_{a_t} \text{vec } M_{t+1} = \sum_i \omega_i \left[\text{vec}(B_{i,t+1}) D_{a_t} p_{i,t+1} + p_{i,t+1} D_{a_t} \text{vec}(B_{i,t+1}) \right].$$

The first term operates through attachment. The second operates through response conditional on attachment. Panel records identify the transition terms directly; repeated cross sections require a model for gross flows and composition.

2.2. Feasible response addition

Let $V = \text{Im } M$ be the current target span and let $W \subseteq \mathbb{R}^q$ contain target directions required by the policy objective. A response addition is a vector $d \in \mathbb{R}^q$ supplied by an eligibility rule, account system, delivery mode, duration rule, or other available instrument. Let $\mathcal{D} = \{d_1, \dots, d_L\}$ be the available dictionary and let $c_\ell > 0$ be the cost of activating d_ℓ . For $S \subseteq \{1, \dots, L\}$, write $D_S = [d_\ell]_{\ell \in S}$ and define

$$(12) \quad k_{\mathcal{D}}(W, V) = \min_{S: W \subseteq V + \text{Im } D_S} |S|,$$

$$(13) \quad C_{\mathcal{D}}(W, V) = \min_{S: W \subseteq V + \text{Im } D_S} \sum_{\ell \in S} c_\ell,$$

with value $+\infty$ when the constraint has no feasible bundle.

THEOREM 2 (Feasible response addition). *Let $\pi_V : \mathbb{R}^q \rightarrow \mathbb{R}^q/V$ be the quotient map and let $d(W, V) = \dim W - \dim(W \cap V)$. A feasible bundle exists exactly when*

$$(14) \quad \pi_V(W) \subseteq \text{span}\{\pi_V(d_\ell) : \ell = 1, \dots, L\}.$$

Every feasible bundle satisfies $|S| \geq d(W, V)$. Equality holds exactly when $d(W, V)$ dictionary vectors have projected span containing $\pi_V(W)$. With unrestricted response vectors and unit activation costs,

$$(15) \quad k_{\mathcal{D}}(W, V) = C_{\mathcal{D}}(W, V) = d(W, V).$$

The dictionary determines the gap between quotient dimension and implementable repair. If available additions project onto a space that excludes one direction in $\pi_V(W)$, repeated activation leaves that target direction outside the response span. If the dictionary spans the missing space through vectors carrying additional components, the minimum

feasible bundle can contain more than $d(W, V)$ elements.

2.3. Option value of a response direction

Let $Z \in \mathbb{R}^d$ be a future target shortfall with finite second moment $\Omega = \mathbb{E}[ZZ']$, and let Q be a positive-definite target-weight matrix. For continuously scalable instruments, a planner with response span V chooses an implementable target change $v \in V$ after observing Z and incurs

$$(16) \quad \ell(V, Z) = \min_{v \in V} \frac{1}{2} (Z - v)' Q (Z - v).$$

Let P_V^Q be the Q -orthogonal projector onto V , and write $R_V^Q = I - P_V^Q$. The ex ante loss is $L(V) = \mathbb{E}[\ell(V, Z)]$.

THEOREM 3 (Response option value). *The expected targeting loss satisfies*

$$(17) \quad L(V) = \frac{1}{2} \text{tr}(Q R_V^Q \Omega).$$

For a response addition d , define $\tilde{d} = R_V^Q d$. If $\tilde{d} \neq 0$, its option value is

$$(18) \quad L(V) - L(V + \text{span}\{d\}) = \frac{1}{2} \frac{\tilde{d}' Q R_V^Q \Omega (R_V^Q)' Q \tilde{d}}{\tilde{d}' Q \tilde{d}}.$$

An addition with $\tilde{d} = 0$ has value zero in equation (16).

Equation (18) assigns value to the component of a delivery option that lies outside the current response span. The target weights and second moments determine which projected direction absorbs more expected loss. Activation costs from equation (12) convert this value into a net-benefit comparison over feasible bundles. Rank and singular values describe the response map under reported units; Q , Ω , and the activation costs complete the planner's ordering.

2.4. Baseline dynamics and intervention dynamics

For a local linear system,

$$(19) \quad \delta s_{t+1} = A \delta s_t + B \delta u_t, \quad \delta y_t = C \delta s_t,$$

the horizon response is

$$(20) \quad M_H = C \begin{bmatrix} A^{H-1} B & A^{H-2} B & \dots & B \end{bmatrix}.$$

Powers of A govern baseline convergence. The image of equation (20) governs policy response capacity. The adoption model below derives a transition for that image from adoption and remaining reach.

THEOREM 4 (Baseline convergence and response rank). *Let $A \in \mathbb{R}^{n \times n}$ satisfy $\rho(A) < 1$, and let $C \in \mathbb{R}^{q \times n}$ have row rank q . Two systems can share A , C , and every baseline trajectory while their horizon response ranks equal zero and q . In particular, $B_0 = 0$ gives $\text{rank } M_H = 0$, while $B_1 = I_n$ gives $\text{rank } M_H = q$ for every $H \geq 1$.*

The two systems generate the same observations under $u_t = 0$. Their intervention laws differ through B . Baseline convergence therefore supplies a law for passive trajectories, while policy experiments and maintained restrictions on B supply information about response rank.

For a nonlinear baseline law $s_{t+1} = F_0(s_t)$, define continuous response strength by $\kappa_H(s) = \|M_H(s)\|_F$.

THEOREM 5 (Convergence to perfect stillness). *Suppose s^* has a neighborhood on which $\|F_0^t(s) - s^*\| \leq K\lambda^t\|s - s^*\|$ for $K < \infty$ and $\lambda \in (0, 1)$. If M_H is locally Lipschitz with constant L and $M_H(s^*) = 0$, then every baseline trajectory in that neighborhood satisfies*

$$(21) \quad \kappa_H(s_t) \leq LK\lambda^t\|s_0 - s^*\|.$$

For a capacity floor $\underline{\kappa} > 0$, the trajectory enters $\{s : \kappa_H(s) < \underline{\kappa}\}$ by every integer date greater than $\log(LK\|s_0 - s^\|/\underline{\kappa})/(-\log \lambda)$.*

3. Instrument Overlap and Future Policy Response Capacity

Two mechanisms shift capacity. A configuration can change the contemporaneous response to an incentive when both instruments operate on the same choice. Earlier adoption can change the response to a later offer through experience and remaining demand. Repeated campaigns carry the resulting transition forward.

3.1. An incentive under an effective cap

Let $x(b) \in \mathbb{R}_+$ be a household's chosen use under incentive status $b \in \{0, 1\}$, with $x(1) \leq x(0)$. A configuration imposes an effective cap $\ell \in \mathbb{R}_+$, producing observed use

$$(22) \quad y(b, \ell) = \min\{x(b), \ell\}.$$

The incentive response under the configuration is $K(\ell) = y(1, \ell) - y(0, \ell)$.

PROPOSITION 2 (Capacity substitution under an effective cap). For $\ell_1 \leq \ell_0$,

$$(23) \quad K(\ell_1) - K(\ell_0) = \text{len}((\ell_1, \ell_0] \cap (x(1), x(0)]) \geq 0, \quad K(\ell) \leq 0.$$

For coupled configured and comparison caps $L^D \leq L^C$ that remain fixed across incentive states, and random potential uses $X(1) \leq X(0)$, the population capacity change is

$$(24) \quad \mathbb{E}[K(L^D) - K(L^C)] = \mathbb{E}[\text{len}((L^D, L^C] \cap (X(1), X(0)])].$$

When the outcome is minutes of use, equation (24) gives the mean use margin covered by the change in effective caps. Compliance and snoozing enter through the distribution of effective caps. If the effective configured cap can depend on incentive status, write it as L_b . The configured incentive response relative to the response under an unrestricted cap obeys

$$(25) \quad \begin{aligned} & \min\{X(1), L_1\} - \min\{X(0), L_0\} - [X(1) - X(0)] \\ & = \text{len}((L_0, \infty) \cap (X(1), X(0)]) + [X(1) - L_0]_+ - [X(1) - L_1]_+. \end{aligned}$$

The first term is common-cap overlap. The second records the change in the effective cap across incentive states. Factorial assignment identifies their sum. The empirical design observes the limit throughout the study and pays the use-reduction bonus during one three-week interval. This timing yields an overlap contrast between the capacity shift during payment and the capacity shift in the surrounding horizons. User-selected tightness remains a post-assignment variable, so the application uses the timing result for the configured bundle and leaves the two terms in equation (25) combined.

3.2. A two-date adoption model

A household has baseline value $v \in \mathbb{R}$ for the product and a net experience effect $g \in \mathbb{R}$ after first adoption. The distribution of (v, g) has marginal density f_v for v . The net effect g combines learning about product quality with the decline in marginal value from owning a durable unit. In period 1 the posted price is p_1 and the subsidy is a . Purchase is

$$(26) \quad D_1(a) = \mathbf{1}\{v \geq c(a)\}, \quad c(a) = p_1 - a.$$

In period 2 the product is offered at price p_2 . The household's response to the offer is

$$(27) \quad D_2(a, p_2) = \mathbf{1}\{v + gD_1(a) \geq p_2\}.$$

The product is unavailable without the second offer, so the period-2 capacity contrast equals

$$(28) \quad K_2(a, p_2) = \mathbb{E}[D_2(a, p_2)].$$

PROPOSITION 3 (Capacity formation by marginal adopters). *Suppose f_v is continuous at $c(a)$ and the conditional distribution of g given v is continuous in v at that point. At values where the displayed indicator is locally constant,*

$$(29) \quad \frac{\partial K_2(a, p_2)}{\partial a} = f_v(c(a)) [\Pr\{g \geq p_2 - c(a) \mid v = c(a)\} - \mathbf{1}\{c(a) \geq p_2\}].$$

An increase in the early subsidy moves the adoption cutoff downward. The bracket in equation (29) compares the marginal household's response to the later offer after early adoption with its response after passing through period 1 without adoption. Experience can move a marginal household above the second purchase threshold. Existing ownership can move it below that threshold. The density $f_v(c(a))$ measures how much population the early subsidy moves into this comparison.

Equation (29) also distinguishes current reach from future response. The current adoption derivative is

$$(30) \quad \frac{\partial \mathbb{E}[D_1(a)]}{\partial a} = f_v(c(a)).$$

Two early subsidies with the same current adoption effect can generate different later capacity effects through the conditional distribution of g among marginal adopters.

3.3. Own and neighborhood exposure

Let a_i be household i 's subsidy and let $q_i(a_{-i})$ be an exposure mapping from neighboring assignments. Baseline value and experience can depend on q_i through product information, disease transmission, or local availability. The two-target response to own and neighborhood exposure is

$$(31) \quad M^A = D_{(a,q)} \begin{bmatrix} \mathbb{E}[D_1] \\ \mathbb{E}[D_2] \end{bmatrix}.$$

The first row measures current adoption. The second row measures the response supplied to the later offer. Column proportionality gives rank one; differing ratios across dates produce a second response direction. The empirical application estimates the discrete-assignment version of equation (31).

3.4. Subsidy choice

Let w_1 be the value of first-period adoption, let w_2 be the value of purchase from the later offer, and let $C(a)$ be the fiscal and administrative cost of the early subsidy. A two-date planner chooses

$$(32) \quad \max_{a \in \mathcal{A}} \{w_1 \mathbb{E}[D_1(a)] + \beta w_2 K_2(a, p_2) - C(a)\}.$$

PROPOSITION 4 (Capacity-adjusted subsidy). *Suppose the objective in equation (32) is differentiable and has an interior maximizer a^D . Then*

$$(33) \quad C'(a^D) = w_1 f_v(c(a^D)) + \beta w_2 \frac{\partial K_2(a^D, p_2)}{\partial a}.$$

The second term is given by equation (29).

The myopic subsidy condition contains the first term on the right side. The dynamic condition also values the later purchase response created or removed at the period-1 cutoff. Fiscal cost, learning, and remaining demand therefore enter the same marginal calculation.

3.5. Repeated reach and perfect stillness

Consider a population divided into three shares at date t : adopters x_t , reachable non-adopters r_t , and an excluded share $e \geq 0$, with $x_t + r_t + e = 1$. A campaign reaches and converts a fraction $\alpha \in (0, 1)$ of the reachable nonadopters. The law of motion is

$$(34) \quad x_{t+1} = x_t + \alpha r_t, \quad r_{t+1} = (1 - \alpha)r_t.$$

The response to another campaign at date t is the discrete capacity contrast

$$(35) \quad K_t = \alpha r_t.$$

THEOREM 6 (Perfect stillness under repeated reach). *Under equations (34)–(35),*

$$(36) \quad r_t = (1 - \alpha)^t r_0, \quad x_t = 1 - e - (1 - \alpha)^t r_0, \quad K_t = \alpha(1 - \alpha)^t r_0.$$

The state converges to $(x^, r^*) = (1 - e, 0)$ and the response to the campaign converges to zero. The fixed point is perfectly still for that campaign. When $e > 0$, the limiting target gap is e , and every addition drawn from the campaign's existing response span leaves that gap unchanged.*

The parameter e separates two fixed-point compositions. At $e = 0$, zero capacity accompanies full coverage. At $e > 0$, zero capacity accompanies a population remaining

outside reach. A delivery method that attaches part of the excluded group adds a response vector to the dictionary in equation (12). The cost of closing the remaining gap is then determined by the feasible-addition problem.

The same path allows heterogeneous response rates. Let $r_0(a)$ denote the initial mass of reachable nonparticipants with conversion probability $a \in (0, 1]$. After t campaigns their mass is $r_t(a) = (1 - a)^t r_0(a)$, and aggregate capacity is

$$(37) \quad K_t = \int_0^1 a(1 - a)^t r_0(a) da.$$

It follows that $K_{t+1} \leq K_t$ because $K_{t+1} - K_t = - \int_0^1 a^2(1 - a)^t r_0(a) da$. The limiting reached share is $1 - e$, while e remains as the target gap. This formulation permits early campaigns to select high-response units and later campaigns to face the surviving low-response types.

4. Identification and Rank Inference

The empirical object involves two interventions. An earlier policy A_t changes attachment, experience, or remaining demand. A later policy B_{t+1} acts through the resulting response state. Let H_t contain pretreatment histories, let L_{t+1} collect variables observed between interventions, and let $Y_{t+1+h}(a, b)$ be the potential target at horizon h .

4.1. Continuous sequential policies

For scalar policies, define the later response and the capacity effect by

$$(38) \quad m_h(a, h_t) = \frac{\partial}{\partial b} \mathbb{E}[Y_{t+1+h}(a, b) \mid H_t = h_t] \Big|_{b=0},$$

$$(39) \quad \theta_h(a, h_t) = \frac{\partial}{\partial a} m_h(a, h_t).$$

The second derivative θ_h measures how the earlier policy changes the response to the later policy.

ASSUMPTION 1 (Sequential assignment and smooth support). *Conditional on H_t , A_t is independent of the potential intermediate variables and outcomes indexed by (a, b) . Conditional on (H_t, A_t, L_{t+1}) , B_{t+1} is independent of the potential outcomes indexed by b . The treatment values used in each comparison have positive conditional density. The outcome regression and the conditional distribution of L_{t+1} are differentiable in the relevant treatment values, and an integrable envelope permits differentiation under the integral.*

PROPOSITION 5 (Sequential identification). *Under Assumption 1 and consistency,*

$$(40) \quad m_h(a, h_t) = \int \frac{\partial}{\partial b} \mathbb{E}[Y_{t+1+h} \mid H_t = h_t, A_t = a, L_{t+1} = \ell, B_{t+1} = b] \Big|_{b=0} dF(\ell \mid h_t, a),$$

and differentiating the right side with respect to a identifies $\theta_h(a, h_t)$.

The formula requires variation in both policies over the reported support. A rollout chosen in response to backlog or shock severity places those assignment variables in H_t or calls for a design that fixes assignment. Panel records can then divide the total capacity effect using the transition terms in Theorem 1.

4.2. Sequential randomization and policy history

At decision point t , let D_{it} and A_{it} indicate mutually exclusive diet and activity notifications with known probabilities p_D and p_A . Let S_{it} be the target state recorded before assignment and define centered current assignments $\tilde{D}_{it} = D_{it} - p_D$ and $\tilde{A}_{it} = A_{it} - p_A$. For a window of w prior decision points, randomized policy history is

$$(41) \quad H_{it}^{(w)} = \sum_{\ell=1}^w (D_{i,t-\ell} + A_{i,t-\ell} - p_D - p_A).$$

The empirical projection is

$$(42) \quad Y_{it} = \alpha_i + \lambda_{d(t)} + \mu_{q(t)} + \beta_D^{(w)} \tilde{D}_{it} + \beta_A^{(w)} \tilde{A}_{it} + \gamma^{(w)} H_{it}^{(w)} + \theta_D^{(w)} \tilde{D}_{it} H_{it}^{(w)} + \theta_A^{(w)} \tilde{A}_{it} H_{it}^{(w)}$$

$$(43) \quad + \delta^{(w)} S_{it} + \psi_D^{(w)} \tilde{D}_{it} S_{it} + \psi_A^{(w)} \tilde{A}_{it} S_{it} + e_{it},$$

where $d(t)$ indexes relative enrollment day and $q(t)$ indexes time of day. The empirical specification also interacts both current assignments with centered linear and quadratic functions of relative enrollment day. Centering makes $\beta_D^{(w)}$ the current diet-notification response at mean randomized history, at the reference pre-state, and at the centered response-time terms. The coefficient $\theta_D^{(w)}$ gives the change in that response generated by one additional assigned notification in the preceding window while the recorded pre-state and response-time profile are held fixed.

PROPOSITION 6 (Randomized-history response). *Suppose current assignment is independent of current potential outcomes conditional on the observed decision history, the assignment probabilities equal $(p_D, p_A, 1 - p_D - p_A)$ on retained decision points, consistency holds, and the conditional current-treatment response is linear in $(H_{it}^{(w)}, S_{it})$ over the reported support. Then weighted centered least squares identifies $(\beta_D^{(w)}, \theta_D^{(w)}, \psi_D^{(w)})$ and the corresponding activity coefficients. With constant assignment probabilities, the weights are common and equation (42) supplies the estimator.*

PROOF. Conditional random assignment makes the centered current-assignment vector orthogonal to every function of the observed history and gives it a positive-definite conditional second-moment matrix when all three assignment cells have positive probability. Multiplying the linear conditional response by the centered assignment vector and by its products with $(H_{it}^{(w)}, S_{it})$ yields the normal equations for the response coefficients. Consistency replaces the potential outcomes in those moments with the observed target. Constant probabilities make the weighting factor common across decision points. $\square$

Micro-randomized trials generate the repeated assignment variation used by equation (42) (Boruvka et al. 2018). Participant clustering preserves dependence created by overlapping histories and repeated outcomes. A cluster-multiplier maximum statistic gives simultaneous coverage across a declared collection of windows, outcomes, and current notification types. A lead-history audit replaces equation (41) with future assignments; sequential randomization places the corresponding interaction at zero in expectation.

For q targets, let $\theta_{Dj}^{(w)}$ and $\theta_{Aj}^{(w)}$ denote the two current-assignment-by-history coefficients for target j , and let Δh_w be the difference between the 90th and 10th percentiles of realized randomized history. The window-specific response-change operator is

$$(44) \quad \Theta^{(w)} = \Delta h_w \begin{pmatrix} \theta_{D1}^{(w)} & \theta_{A1}^{(w)} \\ \vdots & \vdots \\ \theta_{Dq}^{(w)} & \theta_{Aq}^{(w)} \end{pmatrix}.$$

Each column records how randomized history changes the current response to one notification type across the target vector. The empirical singular values divide each row by its common-sample target standard deviation and carry the same transformation into the participant-cluster influence covariance. A Gaussian cluster multiplier supplies a joint operator-norm radius across the five window-specific matrices. Weyl's inequality then gives simultaneous lower endpoints $\max\{0, \sigma_j(\widehat{\Theta}^{(w)}) - c_{1-\alpha}\}$. The lower-rank-uniform procedure described below tests the rank of a selected window, while a maximum statistic over the five matrices protects the rank-zero test against window selection.

4.3. Factorial assignment of intervention and configuration

Let $D \in \{0, 1\}$ assign a policy configuration and let $B \in \{0, 1\}$ assign an intervention whose response depends on that configuration. For target horizon h , define

$$(45) \quad K_h(d) = \mathbb{E}[Y_h(d, 1) - Y_h(d, 0)], \quad \Delta K_h = K_h(1) - K_h(0).$$

PROPOSITION 7 (Factorial capacity contrast). *If $(D, B) \perp \{Y_h(d, b) : d, b \in \{0, 1\}\} \mid G$, every*

assignment cell occurs in each included stratum, and consistency holds, then

$$(46) \quad \Delta K_h = \mathbb{E}_G[\mathbb{E}(Y_h \mid D = 1, B = 1, G) - \mathbb{E}(Y_h \mid D = 1, B = 0, G)]$$

$$(47) \quad - \mathbb{E}_G[\mathbb{E}(Y_h \mid D = 0, B = 1, G) - \mathbb{E}(Y_h \mid D = 0, B = 0, G)].$$

In a regression containing B , D , their interaction, and saturated stratum indicators, the interaction coefficient estimates equation (46). Stacking the coefficients across horizons supplies a capacity-shift vector. Replacing D with mutually exclusive randomized configurations yields one intervention response for each configuration.

Let h^* denote a horizon in which both instruments are active and let $\mathcal{C}$ collect comparison horizons. For weights $w_h \geq 0$ satisfying $\sum_{h \in \mathcal{C}} w_h = 1$, define the overlap capacity shift

$$(48) \quad \Gamma_{h^*, \mathcal{C}} = \Delta K_{h^*} - \sum_{h \in \mathcal{C}} w_h \Delta K_h.$$

PROPOSITION 8 (Factorial identification across policy timing). *Suppose the factorial assignment condition in Proposition 7 holds jointly for the outcome vector indexed by $\{h^*\} \cup \mathcal{C}$. Then equation (48) is identified by the same linear contrast of the horizon-specific interaction coefficients. Its covariance is the corresponding quadratic form in their cross-horizon covariance matrix.*

Random assignment identifies each component of equation (48); linearity identifies the timing contrast. The effective-cap model assigns its mechanism when D changes the cap and B changes the unconstrained choice during h^* . A weekly version can replace the period means and compare the payment weeks with the remaining weeks under the same assignment.

Let $\widehat{\theta}$ stack J horizon interactions, let $\mathcal{A} = \{a_1, \dots, a_L\}$ be a finite collection of timing contrasts, and let $\widehat{\Sigma}$ estimate the joint covariance of $\widehat{\theta}$. Define $s_j^2 = a_j' \widehat{\Sigma} a_j$ and let $c_{1-\alpha}$ be the $1 - \alpha$ quantile of $\max_{j \leq L} |a_j' Z| / s_j$ for $Z \sim N(0, \widehat{\Sigma})$.

PROPOSITION 9 (Finite timing family). *Under joint asymptotic normality and consistent covariance estimation, the intervals*

$$(49) \quad a_j' \widehat{\theta} \pm c_{1-\alpha} s_j, \quad j = 1, \dots, L,$$

have asymptotic simultaneous coverage $1 - \alpha$. The interval indexed by any data-dependent selector taking values in $\{1, \dots, L\}$ inherits that coverage.

The empirical timing audit uses two finite families. The period family compares each of four periods with the equal-weight mean of the other three. The weekly family compares each of ten consecutive three-week windows with the other nine weeks. Gaussian max- t

critical values use the joint HC2 covariance. A conditional assignment audit applies the same four period contrasts to every re-randomization draw. Each draw re-estimates the interaction and its HC2 covariance from the observed outcome vector, then forms studentized statistics. This procedure targets weak null hypotheses for average factorial contrasts while allowing unit-level treatment-effect heterogeneity, following the studentization argument in Wu and Ding (2021). These procedures cover selection among the displayed timing contrasts; the analysis record separately reports when the economic model and each family entered the project. This distinction follows the inference-after-selection problem studied by Andrews et al. (2024).

4.4. Prospective external benchmarks

An external benchmark enters when its trial record predates assignment, its published estimand maps into a configuration or history contrast in this framework, and the manuscript carries the source estimate with its reported uncertainty and multiplicity adjustment. The benchmark retains the source trial's outcome definition, horizon, and analysis family. It supplies evidence about the response mechanism across a separate population and implementation. Cross-trial differences in target units, intervention intensity, and follow-up remain part of the comparison.

4.5. A randomized early policy and a common later offer

The application has a discrete early assignment and a common later offer. Let $A \in \mathcal{A}$ be randomly assigned within block G , let $B = 1$ indicate receipt of the later offer, and let $Y_2(a, b)$ be later purchase. Suppose the offered product is unavailable through the comparison channel when $b = 0$, giving $Y_2(a, 0) = 0$. Define

$$(50) \quad K_2(a) = \mathbb{E}[Y_2(a, 1) - Y_2(a, 0)] = \mathbb{E}[Y_2(a, 1)].$$

PROPOSITION 10 (Randomized capacity effect). *If $A \perp \{Y_2(a, 1) : a \in \mathcal{A}\} \mid G$ and each compared assignment occurs within every included block, then*

$$(51) \quad K_2(a) - K_2(a') = \mathbb{E}_G[\mathbb{E}(Y_2 \mid A = a, G) - \mathbb{E}(Y_2 \mid A = a', G)].$$

The later offer can be common across households because the zero-offer outcome is fixed by product availability. Randomized variation in the earlier assignment identifies how policy history changes the response generated by that offer. A within-block permutation of early assignments supplies finite-sample inference under the corresponding sharp null.

With interference summarized by an exposure mapping $Q_i(A_{-i})$, random assignment

also generates variation in neighbors' policy exposure. The reported estimand then conditions on the chosen mapping and includes own assignment and mapped exposure as separate instruments. Spatial covariance estimates account for overlapping neighborhoods in the outcome equations.

4.6. Response matrices and rank

Let $M \in \mathbb{R}^{q \times m}$ stack target responses to assigned exposures and let $k = \min\{q, m\}$. For $r < k$, the rank null is $H_0^{(r)} : \text{rank}(M) \leq r$. The empirical statistic is

$$(52) \quad T_n(r) = n \sum_{j=r+1}^k \sigma_j^2(\widehat{M}).$$

The singular values use the target and instrument scales stated for each application. The digital matrix divides each target row by its sample outcome standard deviation. Its covariance receives the same row transformation.

The lower-rank-uniform calculation follows the CFA two-step procedure in Chen and Fang (2019). Set $\kappa_n = n^{-1/4}$ and estimate a lower rank r_0 by counting singular values of $\widehat{M}$ above κ_n . A value $r_0 > r$ rejects in the first step. For $r_0 \leq r$, write the singular-vector matrices as U and V , retain the columns associated with indices above r_0 , and draw Z^* from the estimated law of $\sqrt{n} \text{vec}(\widehat{M} - M)$. The bootstrap statistic applies equation (52) to the singular values of $U_0' Z^* V_0$ beginning at index $r - r_0 + 1$. The applications use 49,999 Gaussian draws from the joint coefficient covariance. The digital covariance is HC2 across participants; the bed-net covariance uses the spatial estimator in equation (14). Robin and Smith (2000) give characteristic-root and sequential rank procedures.

When instruments have different exposure schedules, let S contain the known schedule vectors in target units and let R_S project onto their orthogonal complement. The schedule-adjusted matrix is $R_S M$. Its rank records response variation remaining after the scheduled activation profiles are removed. The application reports raw and schedule-adjusted singular values and carries the transformed coefficient covariance into the second rank calculation.

For a two-by-two matrix, the regular rank-one diagnostic also uses $\det M = 0$. At a rank-one matrix, $\nabla \det(M) = \text{vec}(\text{adj}(M)')$ is nonzero, and

$$(53) \quad \text{Var}(\det \widehat{M}) \simeq \nabla \det(M)' \text{Var}(\text{vec} \widehat{M}) \nabla \det(M).$$

The determinant calculation supplies a comparison at regular rank-one points. The two-step calculation covers the lower-rank points contained in $H_0^{(1)}$.

5. Randomized Policy History, Configuration, and Assigned Actions

5.1. Repeated randomized notifications

5.1.1. Design and response estimand

The myBPmyLife intervention followed 287 participants with hypertension for six months and randomized diet and activity notifications at four daily decision points (Golbus et al. 2026). At each point, the probability of a diet notification was 0.125, the probability of an activity notification was 0.125, and the probability of receiving either notification was 0.25. The public core file contains 187,517 randomization points. The analysis uses 186,893 active-phase points with a recorded dice roll. Diet notifications directed participants to the mobile application to find lower-sodium alternatives. The source article classifies 60-minute app engagement as an exploratory post hoc endpoint and reports that diet notifications raise engagement. The present analysis retains that outcome status and asks whether randomized notification history changes the response to another diet notification.

Equation (42) is estimated for histories spanning 4, 12, 28, 56, and 112 preceding decision points, corresponding to about 1, 3, 7, 14, and 28 days. The outcome is an indicator for app use during the 60 minutes after assignment, and the pre-state is app use during the preceding 30 minutes. Every specification includes centered current diet and activity assignments, centered randomized history, both current-assignment-by-history interactions, the pre-state, both current-assignment-by-pre-state interactions, centered linear and quadratic relative-day terms interacted with both current assignments, participant effects, relative-day effects, and time-of-day effects. Thus, the history coefficient compares current policy responses at the same recorded app-use state and response-time controls. Standard errors cluster by participant.

Inference reports two finite families. The displayed family contains the five diet-history coefficients for binary app use. A broader family contains 40 coefficients spanning five history windows, four engagement outcomes, and two current notification types. Gaussian participant-cluster multiplier statistics provide simultaneous coverage. A retrospective multivariate diagnostic arranges each window's eight coefficients as the four-target-by-two-notification operator in equation (44) and applies simultaneous inference across all five matrices.

5.1.2. State transition and response history

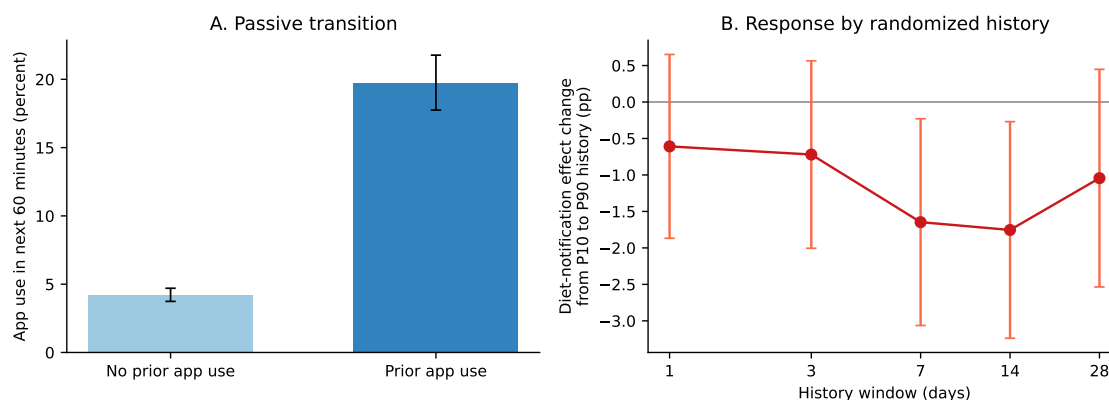


FIGURE 1. App-use transition and response under randomized notification history

Notes. Panel A uses 140,397 active-phase decision points assigned to the 0.75 control condition. Bars report app use during the next 60 minutes by app use during the preceding 30 minutes, with participant-clustered 95 percent intervals. Panel B reports the change in the current diet-notification effect as randomized history moves from its 10th to its 90th percentile. Intervals are simultaneous over the five displayed windows and use 99,999 Gaussian participant-cluster multiplier draws.

At decision points assigned to the control draw, app use during the next 60 minutes equals 4.22 percent after a zero pre-state and 19.76 percent after a one pre-state. Their difference is 0.155 with a participant-clustered standard error of 0.010. Iteration of the corresponding two-state transition gives a fixed point of 5.00 percent. These quantities summarize the passive transition observed in the same randomized decision panel; Table 3 in the appendix reports their uncertainty.

TABLE 1. Current app-use response by randomized notification history

History (days)	Current diet effect at mean history (pp)	Past assignments P10–P90	Response change P10–P90 (pp)	Primary adjusted p	40-coefficient family adjusted p
1	5.31	0–2	-0.61 (0.51)	0.6031	0.9896
3	5.21	1–5	-0.72 (0.52)	0.4681	0.9539
7	5.03	4–10	-1.65 (0.57)	0.0159	0.0878
14	4.84	10–18	-1.75 (0.60)	0.0136	0.0766
28	4.44	22–34	-1.04 (0.60)	0.2668	0.7856

Notes. The sample contains active-phase decision points for the 287-person core cohort. Current diet and activity assignments are centered at their design probabilities of 0.125. History is the number of successful notification assignments during the preceding window centered at 0.25 times the number of decision points. The current effect is evaluated at mean randomized history after a zero app-use pre-state and at the centered time terms. Response change multiplies the history-interaction coefficient by the difference between the 90th and 10th percentiles of realized assignment history; clustered standard errors appear in parentheses. Each row includes both current assignments, history, both current-assignment-by-history interactions, app use during the preceding 30 minutes, both current-assignment-by-pre-state interactions, both current-assignment-by-linear-and-quadratic-day interactions, participant effects, relative-day effects, and time-of-day effects. Primary adjustment covers the five displayed diet-history coefficients. The 40-coefficient adjustment covers five windows, four outcomes, and two current notification types. Gaussian cluster-multiplier inference uses the joint participant-score covariance.

At mean randomized history following a zero pre-state, a current diet notification raises 60-minute app use by 4.44 to 5.31 percentage points across the five specifications. Over the preceding seven days, the 10th and 90th percentiles of assigned notification history are 4 and 10. Moving across that range reduces the current diet-notification effect by 1.65 percentage points, with a clustered standard error of 0.57. The interaction coefficient has $p = 0.0041$, adjustment across the five displayed windows gives $p = 0.0159$, and adjustment across all 40 screened history coefficients gives $p = 0.0878$. Over the preceding 14 days, the corresponding history range is 10 to 18 assignments and the current response falls by 1.75 percentage points, with a standard error of 0.60. Its pointwise, displayed-family, and screened-family values are 0.0035, 0.0136, and 0.0766.

TABLE 2. Policy-response operator across randomized notification history

Target	Diet notification	Activity notification
App use	-1.754 pp	-0.307 pp
App events	-0.211	-0.002
Searches	-0.027	0.001
Low-salt choices	-0.038	0.004
Standardized singular values	0.136, 0.014	
Simultaneous 95% lower endpoint for σ_1	0.017	
14-day rank-at-most-zero p	0.0059	
Five-window adjusted rank-zero p	0.0171	
14-day rank-at-most-one p	0.5774	

Notes. Entries are changes in the current notification response when randomized notification history over the preceding 14 days moves from its 10th percentile (10 assignments) to its 90th percentile (18). The matrix uses the eight 14-day interaction coefficients already included in the 40-coefficient family. Rows are scaled by their common-sample standard deviations for the singular-value calculation. The simultaneous lower endpoint subtracts the 95 percent quantile of the largest operator norm across all five window-specific perturbation matrices, using 99,999 Gaussian participant-cluster multiplier draws. The adjusted rank-zero value also covers all five windows. Lower-rank tests use the Chen–Fang two-step procedure with 49,999 draws. Row scaling changes the metric and leaves algebraic rank unchanged. The source study classifies these engagement targets as exploratory post hoc outcomes.

The row-standardized 14-day operator has singular values 0.136 and 0.014. A simultaneous operator-norm band covering all five history windows gives a 0.017 lower endpoint for its leading singular value. The corresponding lower endpoints are 0.023 at seven days and zero at the other three windows. Rank-zero tests adjusted across the five matrices give $p = 0.012$ at seven days and $p = 0.017$ at 14 days. Replacing Gaussian multipliers with Rademacher participant-cluster multipliers gives adjusted values of 0.0084 and 0.0135 and lower endpoints of 0.024 and 0.019. At 14 days, the lower-rank-uniform tests give $p = 0.0059$ for rank at most zero and $p = 0.577$ for rank at most one. The evidence therefore supports at least one history-sensitive response direction across these engagement targets and leaves a second direction unresolved. The operator construction remains a retrospective diagnostic over outcomes classified as exploratory post hoc by the source study.

The history variable uses assignment success recorded by the randomization engine, so the coefficients are intention-to-treat response changes among retained active-phase decision points. The pre-state interactions allow the current diet-notification effect to vary with current app use, and centered linear and quadratic relative-day interactions allow both current notification effects to evolve over the study. The history interaction therefore records response variation remaining after state and study-time response heterogeneity enter the equation. Replacing the preceding 28 assignments with the following 28 assign-

ments gives a diet interaction of 0.00119 with a standard error of 0.00094 and $p = 0.206$. Sequential assignment gives the future-history audit a zero benchmark.

The distinction is visible across the two panels. Panel A estimates a passive transition for the target state. Panel B estimates a change in the policy-to-target response generated by randomized policy history while the observed pre-state enters the response equation. The result concerns digital engagement among participants retained in the active decision panel. The source outcome, population, and history windows delimit its transport to other policy systems.

5.2. A bonus under randomized app-use limits

5.2.1. Design and estimand

Allcott et al. (2022) recruited 1,933 Android users and recorded use of Facebook, Instagram, Twitter, Snapchat, browsers, and YouTube in five three-week periods. At the start of period 2, participants received random assignment to a temporary bonus that paid for lower use in period 3. They also received independent assignment to a limit-control group or to an app-limit configuration. The configuration assignments used snooze delays of 0, 2, 5, or 20 minutes and a block without snooze. The limit remained available across the study, while cash bonus payments occurred in period 3. The source article estimates habit formation and self-control and reports bonus-by-limit interactions by period in its online appendix.

Let B_i denote bonus assignment and D_i assignment to any limit configuration. The outcome Y_{ih} is average daily FITSBY use in period h . Equation (13) estimates the bonus response $K_h(0) = \beta_h$ under limit control, the response $K_h(1) = \beta_h + \theta_h$ under a limit assignment, and the capacity shift $\Delta K_h = \theta_h$. The period-level overlap estimand applies equation (48) with $h^* = 3$, $\mathcal{C} = \{2, 4, 5\}$, and equal comparison weights. A weekly audit compares weeks 7–9, when bonus payments occur, with weeks 4–6 and 10–15. Every regression includes baseline FITSBY use and the reported randomization-stratum indicators.

The smallest bonus-by-limit cell contains 194 participants. FITSBY use is observed for more than 99 percent of participants in each period. The 80-percent-power MDE for θ_3 is 0.187 period-3 outcome standard deviations. A period-1 balance regression gives a bonus-by-limit coefficient of -2.68 minutes with a standard error of 6.50.

The August 29 analysis record specified period 3 and the periods 2–5 horizon family before the effective-cap model and payment-overlap contrast were added on August 30. The revision retains the period-3 estimate, the joint family test, and the six enforcement-rule estimates. The finite-family audit compares each period with the mean of the other three and compares each consecutive three-week window with the other nine weeks. The payment-overlap result is an exploratory estimate with simultaneous inference over the displayed families. Section 5.3 reports results from trials whose configuration comparisons

and outcomes were registered before treatment assignment.

5.2.2. Response estimates

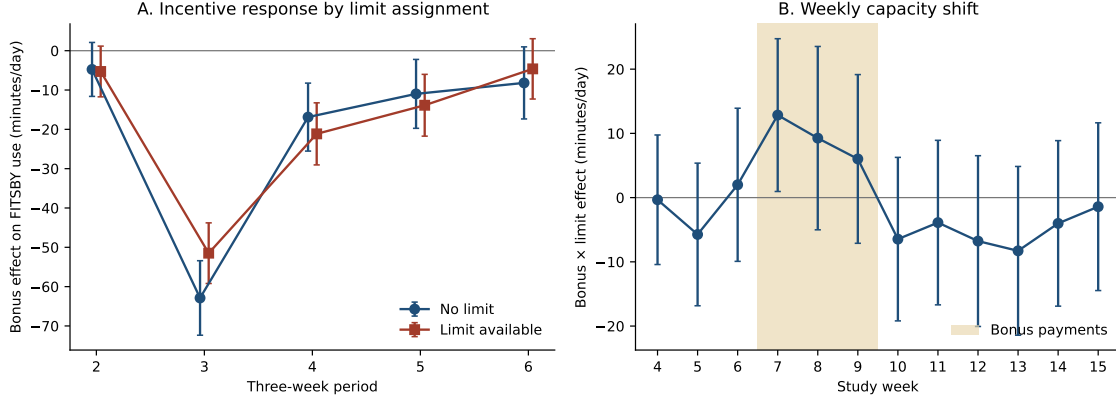


FIGURE 2. Policy overlap and the incentive response

Notes. Panel A reports the estimated bonus effect on average daily FITSBY use by three-week period and binary limit assignment. Panel B reports the weekly bonus-by-limit coefficient from weeks 4 through 15; shading marks the bonus-payment weeks 7 through 9. Bars give pointwise 95 percent HC2 intervals.

Figure 2 follows the incentive response across policy timing. In period 3, the bonus reduces use by 62.88 minutes per day in the limit-control group and by 51.49 minutes under a limit assignment. The period-3 capacity shift is 11.39 minutes with a standard error of 6.24 and a two-sided p -value of 0.068. Its sign agrees with the common-cap term in Proposition 2. The assignment bundles access, enforcement, chosen tightness, and snoozing, so equation (25) gives the interpretation when effective caps vary with bonus status.

The capacity-shift estimates for periods 2–5 are -0.40 , 11.41 , -4.64 , and -3.03 minutes on the common 1,903-person sample. The payment-period shift minus the equal-weight mean of the surrounding shifts is 14.10 minutes, with a standard error of 5.08 and $p = 0.0054$. Across all four one-period-versus-rest contrasts, Gaussian max- t adjustment gives $p = 0.020$ and a simultaneous interval from 1.62 to 26.59 minutes. A 9,999-draw conditional re-randomization follows the recorded assignment sequence and re-estimates the HC2 standard error in every draw. The weak-null values are 0.0059 for the payment contrast and 0.0216 after adjustment over the four-period family. At weekly frequency, the mean shift during weeks 7–9 minus the mean shift in the other nine included weeks is 13.25 minutes, with a standard error of 4.96 and $p = 0.0075$. Across the ten consecutive three-week windows, its adjusted value is 0.047 and its simultaneous interval runs from 0.10 to 26.41 minutes. The adjacent weeks 6–8 window has the largest studentized estimate in that family and an adjusted value of 0.027. The four horizon coefficients have a joint Wald statistic of 9.12 with four degrees of freedom and $p = 0.058$.

TABLE 3. Randomized limit assignment and the response to a temporary use-reduction bonus

	Period 2	Period 3	Period 4	Period 5
Bonus effect without limit	-4.76 (3.50)	-62.88 (4.83)	-16.89 (4.41)	-10.98 (4.47)
Bonus effect with limit	-5.29 (3.29)	-51.49 (3.93)	-21.15 (4.03)	-13.87 (4.01)
Capacity shift: bonus $\times$ limit	-0.52 (4.81)	11.39 (6.24)	-4.26 (5.98)	-2.89 (6.01)
Limit effect	-24.51 (2.29)	-25.62 (2.89)	-21.86 (2.93)	-20.58 (2.89)
Observations	1930	1932	1931	1933

Notes. Each column reports HC2 estimates from a factorial regression of average daily FITSBY use on bonus assignment, limit assignment, their interaction, baseline FITSBY use, and randomization-stratum indicators. Period 3 is the bonus-payment period. Standard errors appear in parentheses.

The limit coefficient ranges from -20.57 to -25.59 minutes per day across periods 2–5. After each horizon is scaled by its outcome standard deviation, the four-by-two matrix of bonus and limit coefficients has raw singular values 0.821 and 0.293. The bonus payment occurs in period 3 and limit access spans periods 2–5. Projecting the response columns away from these two exposure schedules gives singular values 0.094 and 0.024. The lower-rank-uniform schedule-adjusted test gives $p = 0.066$ for rank at most one. The raw rank records the assigned policies together with their timing; the schedule-adjusted calculation leaves the residual response rotation unresolved at the 5 percent level.

The six configuration estimates range from a 44.27-minute to a 71.47-minute period-3 bonus response. Figure 3 in the appendix reports Gaussian max- t intervals across the six estimates; a companion family covers the five comparisons with limit control. Each comparison interval contains zero. The binary configuration contrast supplies the period and timing estimands.

The released author code specifies the same four equations as $Y_h \sim Y_1 + L + B + L \times B + S$. Its assignment program draws bonus status within eight baseline strata, draws limit status within stratum-by-bonus cells, and draws snooze delay within the corresponding delayed-snooze cells. Re-estimation with that equation changes every bonus, limit, and interaction coefficient by less than 2×10^{-10} . The conditional re-randomization preserves treatment counts within these cells and uses observed outcomes with an HC2-studentized statistic. The period-3 weak-null value is 0.069. Studentization permits heterogeneous treatment effects under the weak-null procedure in Wu and Ding (2021).

5.2.3. Choice over assigned actions

The experiment supplies four attainable actions: neither assignment, bonus alone, limit alone, and both assignments. Their target changes relative to the control cell are 0, $\widehat{\beta}$, $\widehat{\delta}$, and $\widehat{\beta} + \widehat{\delta} + \widehat{\theta}$. The planner declares a common daily use-reduction target for periods 2–5 and evaluates equation (4) over these four points. Each horizon is weighted by the inverse of its outcome variance in the 586-person control cell.

TABLE 4. Feasible factorial actions for declared use-reduction targets

Assigned action	15 minutes	30 minutes	50 minutes
None	0.049	0.195	0.543
Bonus	0.124	0.121	0.269
Limit	0.015	0.011	0.158
Bonus and limit	0.272	0.123	0.076
Estimated choice	Limit	Limit	Bonus And Limit
Draw probability of estimated choice	0.998	1.000	0.999

Notes. Each column declares the same daily reduction target for periods 2 through 5. The feasible set contains the four randomized assignment cells. The combined action uses the estimated bonus, limit, and interaction coefficients. Loss is one half of the squared target gap after each row is divided by the corresponding control-cell outcome standard deviation. The final row gives the frequency with which 49,999 Gaussian draws from the joint HC2 coefficient law select the reported action. Costs enter through the boundaries reported in the text.

With zero activation cost, the estimated action is no assignment for targets below 11.61 minutes, limit alone from 11.61 through 41.53 minutes, and both assignments above 41.53 minutes over the reported zero-to-80-minute range. Table 4 evaluates targets of 15, 30, and 50 minutes. The limit is selected in 99.8 percent and 100 percent of the 49,999 coefficient draws at 15 and 30 minutes. Both assignments are selected in 99.9 percent of draws at 50 minutes. Let λ denote dollars per standardized target-loss unit and let c_L be the resource cost of limit implementation. At a 30-minute target, limit assignment beats no assignment when $c_L < 0.184\lambda$ and beats bonus assignment when $c_L < 76.42 + 0.109\lambda$. At a 50-minute target, adding the bonus after the limit requires $\lambda > 929.9$ when the payment is counted as an agency outlay. Mean predicted bonus earnings are \$76.42. Mean bonus valuation is \$64.07, and mean limit valuation is \$4.18 among 1,149 observed responses. A social-welfare objective assigns the transfer incidence and implementation resources before applying the same four-action comparison.

5.3. Registered external benchmarks

Delgado et al. (2024) registered NCT03833219 and randomized 2,020 drivers among control,

feedback, standard incentive, feedback combined with a standard incentive, feedback combined with a reframed incentive, and feedback combined with a doubled incentive. During the 7-week intervention, feedback combined with the standard incentive reduced handheld phone use by 38 seconds per hour relative to control, with a 95 percent interval from 8 to 69 seconds and Holm-adjusted $p = 0.045$. Feedback alone and the standard incentive alone have adjusted values of 0.64 and 0.82. Direct comparisons of the combined arm with these component arms have adjusted values of 0.85 and 0.74. The experiment establishes the combined-arm reduction relative to control and leaves the interaction among the component arms unresolved. During the 6-week follow-up, each active-arm comparison with control has an interval containing zero.

Sachdeva et al. (2026) preregistered AsPredicted 142497 and randomized 149 smart-phone users between self-selected and assigned daily-use goals while holding the incentive rule and average goal fixed. Self-selected goals reduced phone use by 26 additional minutes per day and raised goal achievement by 11 percentage points. The assignment changes the configuration through which the common incentive operates. Together, the two trials show how prospective designs can declare a configuration comparison, its outcome horizon, and its analysis family before assignment. The present digital timing contrast remains labeled by its archived analysis sequence.

5.4. Scope across policy history and repeated reach

The two-stage bed-net pricing experiment in Dupas (2014) provides a history design. Assignment to a Phase 1 price at or below 50 Kenyan shillings raises Phase 1 purchase by 37.2 percentage points and purchase from a common 150-shilling offer one year later by 7.2 percentage points. The within-area randomization values are 0.0001 and 0.0742. A response matrix adds randomized neighborhood exposure; its lower-rank-uniform test of rank at most one gives $p = 0.077$. Appendix B reports the design, matrix, and uncertainty. The later estimate records the total effect of early assignment on response to the common offer.

The StayWell Phase II data record repeated mood-rating prompts in an exposed cohort (Arévalo Avalos et al. 2024). Among 177 participants with at least 50 recorded decision dates, 145 first respond at prompt 1, 19 enter by prompt 11, and prompts 12–50 add zero observed first responses. Thirteen participants remain without an observed response, giving cumulative reach of 92.7 percent. Appendix B reports the fixed-cohort construction. This accounting exercise distinguishes observed entry from the residual cohort gap. Its estimand is repeated reach under common exposure; prompt effects and the composition of the residual gap remain outside that estimand.

6. Conclusion

Policy response capacity records the target change generated by an admissible intervention at a stated horizon. The units attached to a policy system and their conditional behavior determine that response. Joint unit transitions divide changes in the response matrix into entry, exit, and changes among continuing units. Systems with the same attracting baseline law can carry response matrices of rank zero and full row rank. Along a path attracted to a zero-response state, the response norm inherits an exponential upper bound from the baseline convergence rate and the local response map.

Available delivery methods determine which response additions a planner can implement. Quotient dimension gives a lower bound on the number of additions, while the feasible dictionary determines attainment and cost. A finite experiment generates assigned action points and population mixtures of those points. The finite-action proposition gives the condition under which the linear-span relaxation attains the same targeting loss. This condition aligns the planner’s feasible set with the intervention combinations supported by the design.

The myBPmyLife micro-randomized trial measures current notification responses across randomized notification histories. A 14-day history shift from its 10th to its 90th percentile reduces the current app-use response to a diet notification by 1.75 percentage points. The displayed five-window family gives $p = 0.0136$, and the 40-coefficient screened family gives $p = 0.0766$. Across four engagement targets and two notification types, the standardized response-change operators at seven and 14 days have leading singular values 0.142 and 0.136. Simultaneous lower endpoints across all five windows are 0.023 and 0.017, with adjusted rank-zero values of 0.012 and 0.017. The regression carries the observed pre-state, relative study time, and their current-treatment interactions, so the history coefficients measure response variation at the same recorded state and time controls. A future-history interaction gives $p = 0.206$. The passive transition and current randomized response come from the same 287-person decision panel. The exploratory classification of the source outcomes carries through to the operator calculation.

The factorial digital-use experiment measures the response to a temporary bonus under randomized app-limit configurations. The payment-period interaction equals 11.4 minutes per day with $p = 0.068$. The exploratory payment-period-versus-rest contrast equals 14.1 minutes. Simultaneous Gaussian and weak-null assignment procedures give family values of 0.020 and 0.0216. The analysis record dates the model-guided contrast after the period estimates, while registered external trials supply prospectively declared configuration comparisons.

The raw bonus-and-limit response matrix combines intervention effects with their assigned exposure schedules. Projection away from those schedules reduces the singular values from 0.821 and 0.293 to 0.094 and 0.024. The schedule-adjusted test of rank at most

one gives $p = 0.066$. The resulting claim concerns the assigned time profiles; a second residual response direction awaits a design with independent timing variation.

The estimated policy choice evaluates the control assignment, bonus alone, limit alone, and both assignments. Under zero activation cost, limit assignment minimizes target loss from 11.61 through 41.53 minutes of common daily reduction, and the combined assignment minimizes loss above 41.53 minutes over the reported range. Bonus outlays and limit implementation resources enter through stated cost boundaries. This calculation uses treatment combinations generated by the experiment.

The bed-net experiment records an earlier-assignment effect on response to a later common offer. The StayWell exercise records first responses within a cohort observed for 50 prompts. Its residual gap combines exclusion, low response probability, attrition, and missing measurement. These scope applications place policy history and repeated reach beside the randomized configuration design while preserving the estimand supported by each dataset.

Perfect stillness occurs when the baseline path converges to a state whose stated intervention generates zero target change. The limit can follow population coverage or exhaustion of the delivery route among the units that remain. The target gap, intervention set, and response path determine which case applies.

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